Purushothaman Natarajan

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Summary

Senior AI Systems Engineer with 5+ years architecting and owning production-grade GenAI, Agentic AI, and Computer Vision platforms across enterprise domains. Specialized in designing hybrid RAG systems, multi-agent orchestration pipelines, and cost-governed LLM deployments — with measurable outcomes including 91% reduction in manual processing load and 85% reduction in document review effort across high-volume enterprise workflows. Brings end-to-end ownership across system design, retrieval strategy, prompt optimization, cost governance, and post-deployment observability. DRDO-funded researcher with three peer-reviewed publications and open-source contributions. Experienced in stakeholder alignment, async cross-functional delivery, and translating ambiguous business requirements into robust, monitored production systems.

Skills

LLM Systems & GenAI: GPT-4, Claude, LLaMA, Mistral, Qwen; Prompt Engineering, DSPy Optimization, LoRA Fine-tuning, Instruction Tuning, Rule-based & Character-level Guardrails (enterprise), Tool Calling, Model Routing, Token Budget Governance
Agentic AI & Orchestration: LangGraph (stateful deterministic workflows), LangChain, DSPy (prompt optimization), AutoGen, CrewAI; Multi-Agent Design, A2A Communication, Tool Orchestration
RAG & Retrieval: Hybrid BM25 + Dense Retrieval, Cross-Encoder Reranking, Semantic/Recursive Chunking, FAISS, Pinecone, Milvus, USearch; RAGAS, TruLens Evaluation
Observability & MLOps: Langfuse, MLflow, Arize Phoenix; CI/CD Pipelines, Prompt Regression Testing, Model Drift Monitoring, Latency Dashboards, Docker, Kubernetes, FastAPI
Computer Vision & Document AI: YOLO, SAM, SegFormer, DeepLabV3+, ViT-STR, MGP-STR, SwinIR, BSRGAN; Object Detection, Segmentation, OCR, Edge Inference, ONNX
ML & Deep Learning: Transformers, CNNs, RNNs, LSTM, Diffusion Models, GANs, SimCLR, RL (PPO, DPO), LIME, SHAP, Grad-CAM
Cloud & Infrastructure: Azure (AI Foundry, ML), GCP
NLP: NER, Text Classification, Information Retrieval, Topic Modeling, PII Detection
Databases: MySQL, PostgreSQL, Neo4j, LMDB, Vector DBs
Programming: Python (primary), SQL, C++ (basic), JavaScript (basic)
Security & AI Risk: CVE Analysis, Threat Intelligence Ingestion, Risk Scoring, Secure ML Deployment

Experience

Data Scientist (AI/ML & Agentic Systems), Blackstraw AI – Chennai, IN — Nov 2024 – Present

- Architected a production-grade hybrid RAG + multi-agent platform spanning document intelligence, vulnerability research, and product data pipelines — owning system design, retrieval strategy, model selection, and deployment — reducing a team of 5's manual processing load from ∼110 hrs/week to under 10 hrs/week (91% reduction across 15K+ monthly document queries).
- Fine-tuned Qwen-72B for item coding classification via LoRA instruction tuning on domain-specific classification guidelines — replacing the DSPy prompt optimization approach and achieving **50–60% workflow automation** at >**95% accuracy** across 3,600+ item categories; selected LangGraph for stateful deterministic orchestration of the fine-tuned model's multi-step decision pipeline.
- Designed recursive + semantic chunking strategy optimized for long enterprise PDFs; benchmarked OpenAI, BGE, and E5 embedding models for domain fit; implemented hybrid BM25 + dense retrieval with cross-encoder reranking; evaluated pipeline quality via RAGAS on a curated 300-sample golden set (faithfulness, answer relevancy, context precision).
- Reduced LLM inference and search costs by ∼35% through prompt compression, aggressive response caching, and a model routing layer dynamically switching between GPT-4 and smaller SLM fallbacks based on task complexity; introduced per-workflow token budget governance.
- Built production observability stack using Langfuse trace logging and custom evaluation harnesses —

monitoring agent tool call reliability, hallucination rates, latency distributions, and prompt regression across deployed agentic systems.

- Led technical delivery across a cross-functional team of 3–4 engineers: unblocked architecture decisions on retrieval stack design and agent state management; defined team coding standards and LangGraph workflow conventions; ran sprint planning and review cycles; mentored junior engineers on prompt optimization and agentic debugging patterns; co-defined deployment SLAs with operations stakeholders; owned production incident resolution end-to-end. Participated in technical hiring as an interviewer, contributing to candidate evaluation and hiring decisions for AI engineering roles.
- Architected OCR and document intelligence stack (SegFormer, DeepLabV3+, ViT-STR, SwinIR/BSRGAN super-resolution) — cutting manual review from ∼8 hrs/day across 1,000+ daily documents to under 1.5 hrs (∼85% reduction); improved OCR extraction F1 from 74% to 91% on complex scanned receipts.
- Built CI/CD pipelines for model versioning and rollback; implemented prompt regression testing and experiment tracking via MLflow; monitored model drift and inference latency across production deployments.
- Implemented automated PII detection and redaction pipelines across documents and images using NLP + embedding similarity matching.
- Designed multi-agent CrewAI workflows for automated codebase migration (Oracle→MySQL, Java→Python) with validation and tool orchestration stages.

Research Fellow (Machine Learning & eXplainable AI), SRMIST – Chennai, IN Sept 2023 – Oct 2024

- Selected as DRDO-funded PhD research fellow focused on explainable and reliable AI systems for defense-grade sonar imaging applications.
- Developed deep learning models for underwater sonar object detection and classification under low-signal and noisy sensing conditions.
- Built explainability-first ML pipelines integrating LIME, SHAP, and Grad-CAM to produce interpretable predictions for mission-critical AI systems.
- Delivered self-explainable AI models validated and tested by NPOL and DRDO teams for operational underwater surveillance scenarios.

Machine Learning Associate, Amazon – Chennai, IN Jul 2022 – May 2023

- Improved annotation and data quality pipelines for Alexa, Ring, and Halo ML systems using NLP preprocessing and normalization workflows.
- Standardized labeling and validation processes, reducing downstream model noise and improving training dataset consistency.
- Built performance dashboards and QA metrics tracking that enabled faster feedback cycles and process optimization.

Customer Support Executive, Amazon – Coimbatore, IN Aug 2021 – Nov 2021

- Managed customer and seller queries, refunds, and returns, achieving top performer status for two consecutive months.

Education

SRM University, PhD in Computer Science Feb 2024 – Nov 2024

- **Coursework:** eXplainable AI, Artificial Intelligence, and Comparison of Learning Algorithms.
- *Transitioned from doctoral research to full-time applied AI engineering following successful delivery of DRDO-funded defense AI initiatives and emerging leadership opportunities in industry.*

BITS, Pilani, M.Tech in Data Science Sept 2022 – Sept 2024

- **Coursework:** Data Science, Applied Machine Learning, Deep Learning, Natural Language Processing, Information Retrieval, Artificial and Computational Intelligence.

Anna University, B.E in Mechanical Engineering Aug 2015 – Nov 2020

- **Coursework:** Kinematics of Machinery, Thermodynamics, Manufacturing Technology, Internal Combustion Engines, Design and Development.
- *Self-directed transition to AI/ML from 2021 through graduate study, independent research, and freelance delivery.*

Publications

Synth-SONAR: Sonar Image Synthesis with Enhanced Diversity and Realism via Dual Diffusion Models and GPT Prompting Sept 2024
Cited by 3

Purushothaman Natarajan, Kamal Basha, Athira Nambiar

VALE: A Multimodal Visual and Language Explanation Framework for Image Classifiers using eXplainable AI and Language Models June 2024
Cited by 10

Purushothaman Natarajan, Athira Nambiar

Underwater SONAR Image Classification and Analysis using LIME-based Explainable Artificial Intelligence Dec 2023
Cited by 5

Purushothaman Natarajan, Athira Nambiar

Projects

Autonomous Vulnerability Intelligence Platform Enterprise

- Designed and owned a production autonomous vulnerability intelligence platform — architecting the ingestion, reasoning, and remediation layers end-to-end across continuously updated CVE and security intelligence sources.
- Implemented hybrid BM25 + dense retrieval with cross-encoder reranking for precise CVE context retrieval; selected BGE-large as embedding backbone after benchmarking against OpenAI and E5 across domain-specific recall; evaluated pipeline using RAGAS on a 300-sample golden set.
- Built hierarchical multi-agent orchestration: triage agent → exploitability reasoning agent → remediation packaging agent, with state persistence across LangGraph workflow nodes.
- Introduced token budget governance and GPT-4/SLM routing to reduce per-query inference cost by ∼30% while maintaining output quality on high-priority CVEs.
- Monitored agent reliability in production via Langfuse trace logging — tracking tool call success rates, reasoning consistency, and hallucination frequency across live CVE ingestion cycles.
- Engineered unhappy-path resilience: implemented per-node retry logic with exponential backoff, fallback chains routing to simpler SLMs on tool call failure, and graceful degradation modes that return partial structured output rather than silent failure — reducing pipeline crash rate from ∼18% to under 3% in production.
- **Tools:** Python, Azure workflows, LLM Agents, AI Foundry, Hybrid RAG, Vector DB, RAGAS, Langfuse, Workflow Orchestration.

Retail Vision Intelligence — Detection & Large-Scale Classification on Edge Devices Enterprise

- Architected edge-deployable computer vision system for retail shelf monitoring — owning model selection, inference optimization, and deployment pipeline across complex in-store environments.
- Designed a vector search based classification system supporting **200K+ product labels** — chosen over traditional softmax classification for scalability and zero-shot extensibility; improved top-5 retrieval accuracy by 14% over baseline using SimCLR-pretrained embeddings.
- Optimized inference pipeline latency from ∼340ms to under 80ms on edge devices through model quantization and ONNX export, enabling real-time shelf analytics.
- **Tools:** YOLO, Image Embeddings, Vector Search, OpenCV, PyTorch, ONNX, Edge AI Optimization.

Hierarchical LLM Decision Automation Engine for Item Coding Enterprise

- Designed a hierarchical LLM reasoning engine to replace manual item coding workflows across 3,600+ categories — architecting the playbook generation, decision tree structuring, and validation layers end-to-end.
- Selected DSPy for systematic prompt optimization over manual engineering — achieved **70% automation** of previously manual item coding, reducing average coding time from ∼4 min/item to under 75 seconds at scale.
- Built image-text noise filtering and rule-based validation layers improving pipeline robustness across noisy product metadata inputs.
- **Tools:** DSPy, Python, LLMs, Hierarchical Reasoning, Image-Text Processing, Rule-based Validation.

Agentic Product Packaging Intelligence System Enterprise

- Architected a search-augmented LLM classification system to automate standards-based product packaging code mapping — replacing a fully manual analyst workflow across thousands of SKUs.
- Selected DSPy for systematic prompt optimization over manual engineering; combined SERP-retrieved product context with structured LLM reasoning to handle ambiguous and incomplete product metadata at scale.
- Designed a multi-stage pipeline: SERP retrieval $\rightarrow$ attribute extraction $\rightarrow$ DSPy-optimized classification $\rightarrow$ rule-based character-level guardrail validation, ensuring output codes conformed to enterprise packaging standards before write-back.
- Reduced manual packaging code lookup and enrichment time from $\sim$6 min/SKU to under 45 seconds, enabling processing of 500+ SKUs/day with a 2-person team.
- Built confidence-scored output layer routing low-certainty predictions to human review, maintaining $>$95% precision on auto-approved codes.
- **Tools:** DSPy, Python, SERP API, LLM Reasoning, NLP Pipelines, Rule-based Validation.

Logistics Asset Detection & Cross-Location Package Similarity Tracking Enterprise

- Built computer vision pipelines to detect logistics assets including packages, cardboard boxes, pallets, and forklifts across warehouse environments.
- Implemented embedding-based similarity matching to determine whether packages observed across different warehouse locations correspond to the same physical item.
- Improved clustering quality and reduced misclassification using SimCLR-based representation learning and anomaly detection methods.
- **Tools:** YOLO, SimCLR, CNNs, Embedding Similarity Search, OpenCV, PyTorch.

Enterprise Document Intelligence Platform — OCR Receipts Enterprise

- Architected a production document AI pipeline converting large-scale scanned receipt and retail documents into structured analytical datasets — owning model selection, preprocessing stack, and post-processing validation.
- Improved OCR extraction F1 from 74% to 91% on a 2,000-document held-out test set of complex multi-column retail receipts, using FFT-former deblurring + SwinIR/BSRGAN super-resolution as preprocessing stages ahead of ViT-STR.
- Reduced super-resolution inference latency from $\sim$16s to 2–3s through INT8 quantization of the SwinIR model and tile-based parallel batch processing — without measurable degradation in downstream OCR quality.
- Built layout-aware post-processing using reading-order detection and structured field extraction, reducing downstream data cleaning effort by $\sim$60%.
- **Tools:** ViT-STR, SwinIR, BSRGAN, FFT-former, ONNX, PyTorch, NLP, Data Post-processing.

Agentic Item Description Expansion using LangChain & LangGraph Enterprise

- Built a multi-agent pipeline to expand abbreviated product titles into rich, structured item descriptions for large-scale e-commerce catalogs.
- Designed LangGraph-based multi-step workflows integrating retrieval, reasoning, and intelligent generation; used semantic chunking with sliding-window overlap and embedding-based reranking to improve retrieved context relevance.
- Implemented LangChain tool-calling agents for product metadata retrieval, attribute extraction, and validation.
- **Tools:** LangChain, LangGraph, LLM Agents, lmdb, Python, Prompt Engineering, Retrieval Pipelines.

DL-Studio — Unified AI Deep Learning Lab Open Source

- Unified deep learning experimentation platform for regression analysis, automated data preprocessing, and competitive benchmarking across 14+ traditional ML and deep learning architectures.
- Repo: github.com/Purushothaman-natarajan/doctane
- **Tools:** Python, PyTorch, Scikit-learn, Streamlit.

Doctane — Modular OCR & Document AI Framework Open Source

- Built an open-source OCR and document understanding repository integrating state-of-the-art models.

- Implemented modular pipelines for text extraction, preprocessing, and document intelligence tasks.
- Repo: github.com/Purushothaman-natarajan/doctane
- **Tools:** Python, Transformers, PyTorch, NLTK, Streamlit.

Explainable Remote Sensing Vision Framework Research

- Built a transfer learning-based remote sensing scene classifier with model explainability.
- Applied LIME and Grad-CAM to generate visual explanations for model predictions.
- Repo: github.com/Purushothaman-natarajan/eXplainable-AI-for-Image-Classification-on-Remote-Sensing
- **Tools:** Python, TensorFlow, Scikit-learn, LIME, Grad-CAM, Gradio.

Additional Experience and Awards

Independent AI Systems Consultant (Upwork & LinkedIn, 2022–2024): Independently designed and delivered end-to-end AI and ML systems for international clients across NLP, computer vision, and data pipeline domains — projects valued between $10K–$20K. Demonstrates that production delivery depth transfers beyond a single employer context.

Instructor, BrightNext Academy (2023–2024): Taught Machine Learning and Deep Learning courses to over 100 students.

Business Analyst, IIFL (Oct 2019 – May 2021): Managed daily client transactions worth $10M, liaised between clients and the trading desk, and provided strategic market insights.

Third Prize, Innovation & Design on Remote Sensing Data by Hack2Skill: Recognized for designing a synchronized research and production-ready dashboard for an explainable image classifier, competing against 100+ entries.

Licenses & Certifications

Udacity: Deep Learning, Computer Vision, Generative AI Nanodegrees.

DeepLearning.AI: Retrieval Augmented Generation (RAG), Reliable AI using Guardrails.

Codecademy: BI Dashboards with Power BI & Tableau, Business Intelligence Data Analyst, SQL for Marketers and Product Managers.

LinkedIn: Advanced SQL, AI Foundations (ANN, CNN, R-CNN, RNN, LSTM, GNN & Transformers), Advanced NLP with Python, GANs and Diffusion Models with TensorFlow and PyTorch, Transfer Learning Using PyTorch, Deep Learning for Computer Vision Applications.